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Disease Prediction and Ayurvedic Guidance Applying Machine Learning

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***Abstract*—Awareness in integrated healing approaches such as Ayurveda with modern technology like machine learning (ML) has developed due to the increasing availability, demand and need for provision of healthcare services. The initiative called "Disease Prediction and Ayurvedic Guidance" seeks to address diseases like stomach upset and headaches through symptom reports that users have made in order to prevent the occurrence. At the point when a health issue is identified beforehand, the platform will provide tailored therapy suggestions. The project utilizes decision tree algorithms to create an efficient and easy to-understand model for accurately diagnosing diseases by studying different symptom patterns. Analysis of models showed that the Decision Tree model performed well with an accuracy of 94.74%, recall rate of 85.71% and an F1 Score of 87.88%. By integrating machine learning with Ayurvedic principles this project introduces an approach to modern healthcare by combining the predictive power of decision trees with the holistic and preventive aspects of Ayurvedic medicine.**

***Index Terms*—Machine Learning, Ayurveda, Disease Prediction, Decision Tree Classifier, Personalized Medicine.**

I. Introduction

Ayurveda, which means "the science of life," is an ancient, well-developed system of medicine that has its roots in over 3,000 years ago in India [1]. Essentially, it aims for health and well-being through the balance of the mind, body, and spirit which is in contrast with Western medicine which has been diagnosis and treatment of ailments which unfortunately is superficial; Ayurveda finds the causes of diseases and maintains the balance of the natural processes of the body. Reshaping health from purely clinical practice to integration of many artistic practices including leveraging traditional Chinese medicine and it is all thanks to the rapid developments in artificial intelligence.

In recent times, the concept of personalized medicine has reached new heights primarily owing to the huge availability of digital health records and machine learning techniques. The idea behind personalized medicine is to offer health care that is tailored to individual patients, with the belief that a person's response to medical treatments can be anticipated and, in any case, is always more effective than the use of any medical treatment in its generic sense. This vision of providing more tailored medicines resonates well with the Ayurveda-centric approach where the treatment modalities fit the individual's constitution (Prakriti) [2], daily habits, and other factors. Although this is the case, the incorporation of traditional Ayurvedic practices with recent technologies has been inadequate, limiting their wider usage in modern medical science.

"Disease Prediction and Ayurvedic Guidance" helps to introduce machine learning models to be integrated with Ayurvedic treatments to provide modern healthcare assistance. This involves creating a web-based application that uses machine learning techniques, particularly decision tree algorithms, to predict common diseases based on symptoms provided by users.[3]

After forecasting a condition, the application offers individualized Ayurvedic treatment suggestions based on treatment free from side effects[4] and practices of Ayurveda which is natural and entire.

This project, by introducing a new approach toward personalized health care through the concentration of practical application of machine learning within an Ayurvedic framework, aims to bring together universal principles of traditional medicine with contemporary technology for a tool that is advanced yet grounded in established principles. The web application developed through this project represents a significant advancement in making personalized health care accessible and effective for diverse populations.

II. Literature Survey

The combination of machine learning (ML) and Ayurveda integration application is a new fucture in medicine that has a high level of promise for the delivery of effective healthcare solutions that are user-friendly and personalized for the users.

Machine learning the application of AI into occupation based applicability with Ayurveda. – Sanika Mohagaonkar, Anmol Rawlani, Ankur Saxena. Islamabad, 2019, March 13-15, conference proceedings. As shown in [5], their research

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focuses on the application of machine learning algorithms in Ayurvedic medicine employing decision tree approaches such as Scikit-learn to categorize Ayurvedic medicines successfully.

In the year 2021, Manjula and AnandaRaj wrote the paper which delved into how ayurveda could benefit from machine learning technologies in the area of diagnosis of diseases. They highlight the importance of lifestyle in health as well as other chronic diseases due to the imbalance of doshas [6].

Khan et al. (2024)'s AI-based Ayurvedic recommendation system addresses the personalization of treatment using Machine Learning. Apart from the patient's Vikruti, the system analyzes various patient attributes including their lifestyle and applies algorithms to recommend appropriate interventions. That improves user accessibility and assists the practitioners through evidence based information [7].

The 2023 paper by deshmukh and khemdchandani discusses pulse analysis as a method of diagnosis in Ayurveda, particularly within the framework of vata, pitta and kapha dosha imbalances. The research seeks to assess the correlation between Nadi Pariksha [8] and the outputs of a peripheral pulse analyzer – thereby creating techniques to advance Ayurvedic treatment with the use of contemporary tools. The study also highlights that the balance of these doshas determines both physiological and psychological characteristics.

Keniya et al. were able to create a system that can predict diseases with the weighted KNN model achieving 93.5% accuracy [9], based on some given age, gender, and symptoms. This system assists in quick diagnosis but could be data constrained.

According to Bujunuri Harish Reddy and M. Sandhiya (2023), the Decision Tree Algorithm achieved 96.10% accuracy in predicting diseases from symptoms [10], significantly surpassing the Gradient Boost Classifier, which achieved 75% accuracy.

Madaan and Goyal investigate the use of advanced machine learning techniques in Ayurvedic constitution predicting and concentrating primarily on the doshas – Vata, Pitta, and Kapha. They conducted a study with a questionnaire that had 28 features among 807 respondents and adopted a balanced sample of 405 subjects [11]. Among other methods, CatBoost achieved the best result with the precision and accuracy of 0.96 and 0.95 respectively. The study also demonstrates the benefits of the use of ensemble learning techniques in improving the diagnostics based on ayurveda.

Sethi et al. in their 2019 paper have proposed a disease prediction mobile application based on userreported symptoms. The authors, though aware of the day-by-day rising role of mobile technology in healthcare, further elaborate that the current systems are mostly limited to medical practitioners or not portable [12]. This would have been an application for disease diagnosis through symptom analysis with parameters like age, gender, and symptoms to serve the purpose better with accurate predictions.The application provides a suggestion for nearby hospitals in order to consult further, so that having smooth and easily accessible experience with digital healthcare services.

Dahiwade et al. created a model for the early prediction of disease by using machine learning applied in practicing doctors in their accurate diagnosis of conditions based on symptoms. They claim that 'the practice is shown as an exploration to get the habits of looking around in the medical data to uncover hidden patterns'. The K-Nearest Neighbor (KNN) and Convolutional Neural Network (CNN) algorithms are used for disease prediction [13], and it has been mentioned that CNN achieved an accuracy of 84.5%, which out performs KNN. The system also estimates the diseases that are relatively related risks, and it provides information regarding risk levels that are either lower or greater mainly due to patient habits and checkup details.

Swathi et al. explored the integration of Ayurveda with transdisciplinary approaches toward a framework of evidence-based personalized medicine. They emphasized tridoshas (Vata, Pitta, Kapha) and Prakriti in terms of a understanding of disease susceptibility and treatment. The paper could be seen reverse pharmacology as an explanation of the mechanisms of Ayurvedic medicine [14], as well as they touched the facet of role of epigenetics for relating lifestyle and dietary aspects to health outcomes. By examining the microbiome's connection to Prakriti, the authors suggest that these methodologies can effectively bridge traditional and modern medicine, providing scientific support for Ayurvedic practices.

By integrating decision trees for disease prediction with Ayurvedic recommendations in a digital format, this project builds upon current research and advances it further. Unlike prior studies that focus solely on generic machine learning models, this work uniquely incorporates Ayurvedic concepts, including individual constitutions (Prakriti), to create personalized treatment plans. The decision tree algorithm predicts diseases based on user reported symptoms, offering a clear and understandable model, which is crucial for healthcare settings where transparency is essential. .

III. DATASET DESCRIPTION

In this study, we seek to use the two diverse datasets available at open-source platform to tackle the problem of classifying diseases, as well as making recommendations on Ayurvedic treatments.

A. Disease Dataset

The disease dataset has all possible symptoms and it has over a hundred different symptoms. Some of the main symptoms including itchy skin, rashes, nodular skin lesions, constant bouts of sneezes, running nose, cough with shivers, and fever are listed in binary. This makes it easy to represent how symptoms relate with the disease and its helpful in the creation of classifiers. To enhance the comprehension level of how the symptoms and disease categories relate, we include below a heat map that shows the symptom distribution across diseases. Also, in the dataset, the number of occurrence of each symptoms can also be presented in the form of bar graph to show its occurrence and importance in the data.

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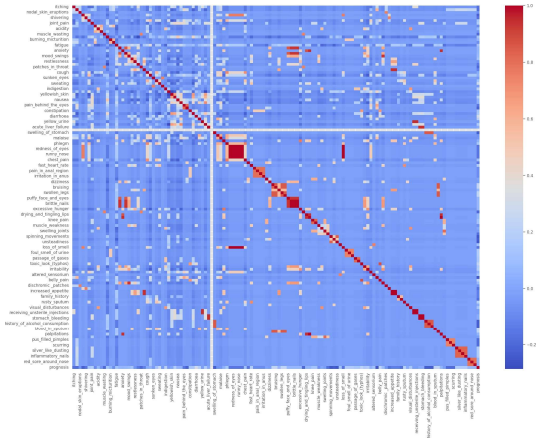

Fig. 1. Correlation heatmap showing relationships between different symptoms and features.

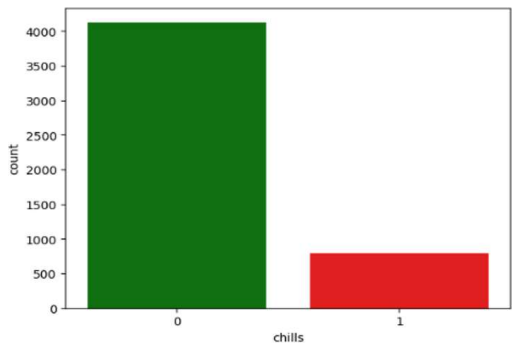

Fig. 2. Symptom Frequency Overview: Analyzing Key Symptoms

B. *Ayurvedic Dataset*

The Ayurvedic dataset presently contains details of few diseases, it has a good span across various demographics. The

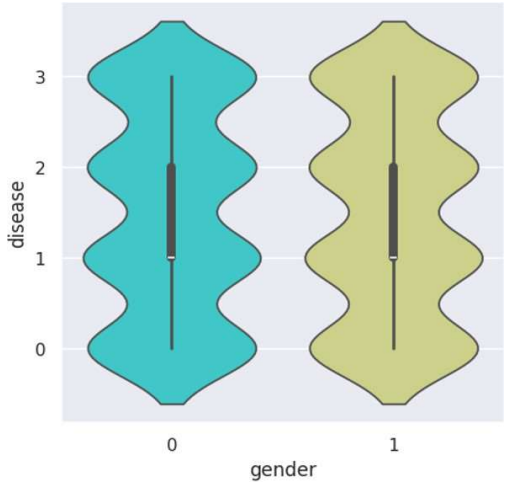

Fig. 3. Exploring Disease Distribution by Gender Using a Violin Plot

data has columns for Disease, Age, Sex, Degree of illness severity, and Suggested Ayurveda medication prescribed, all in binary encodings. It helps us study the demographics and how these treatments differ according to them. Gender distribution among the patients is shown below in the form of a violin plot.

IV. Proposed Work

The aim is to predict the disease and to provide recommendation of Ayurvedic medicine, this is a task that would go a very long way to improve results in health service by using complex machine learning techniques.

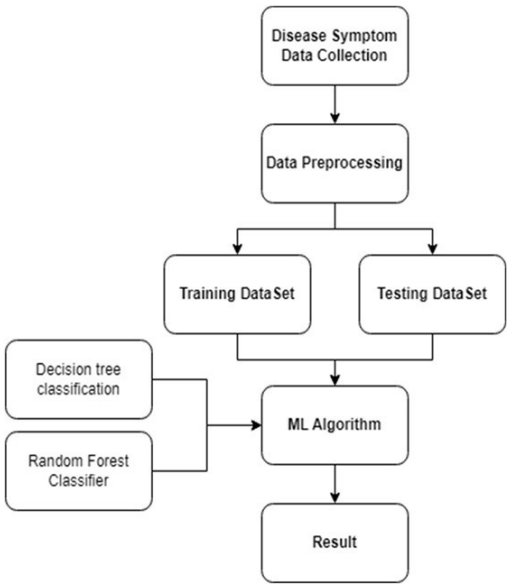

Fig. 4. Proposed Architectural Design: Overview

This will comprise designing and testing of machine learning models, which classify and recommend Ayurvedic medicines for a given sample dataset.

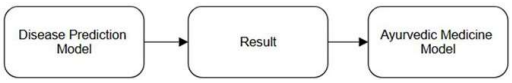

Fig. 5. Flow Diagram

A. *Data Preparation*

1) *Data Cleaning:*

- **Handling Missing Values:** Unresolved missing data may lead to bias in the model and negatively affect predictive power. Different forms of missing data can suggest different reasons, for instance, uncompleted surveys, malfunctioning sensors, or destruction of data. This circumstantial factor reinforces the fact further that the learning of the model will not rely on any imputations of data and assumptions about symptom patterns in cases where they do not exist.
- **Duplicate Entries:** The existence of repeated observations may unfairly bias the model's learning towards

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certain patterns and may result in overfitting to some instances. Upon further investigation of the dataset, it was established that there were no duplicate records, and this has helped in maintaining data integrity while minimizing the chances of repetitions.

- **Data Type Validation:** Training each column's data type allows us to make sure every feature works perfectly in the model. As an example, the symptom columns should be in binary (or absent), while the feature should be categorical.

2) Outlier Detection: Often, atypical instances can skew the model's understanding of common symptom trends resulting in prediction errors in most cases. In the symptom columns, outliers were found using boxplot visualizations. After scrutinizing the data, it was found that there were no apparent outliers, which suggests that the symptom data is evenly distributed throughout the dataset. This step also helps to reduce the likelihood of the model overweighting infrequent values or cases, leading to better prediction accuracy on typical diseases.

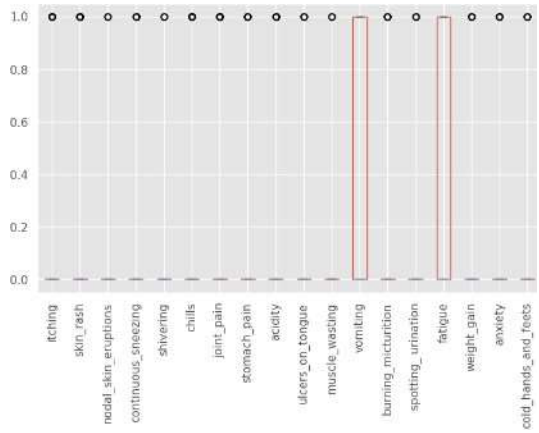

Fig. 6. Analyzing Outliers using Box Plot Visualization

B. Feature Engineering

1) Feature Selection and Standardization::

- **Feature Selection:** There are 133 symptom columns present such that it becomes very necessary to filter out and keep only those features which are most relevant for disease prediction. Too many features cause negative effects as they cause overfitting, which is more or less memorizing things.
- **Standardization:** In order to allow for the equal effect of each feature on the model, all the features were standardized using the StandardScaler to a certain scale. Therefore, as all the features are scaled, we limit the biases' effect on the model performance, allowing the method to consider each of the symptoms equally.

2) Training and Testing Splits: For an objective assessment of the performance of the developed model, the dataset was divided into training and testing datasets. Such separation of data allows training the model on one subset of the data and evaluating its performance on another subset of the data, thus reflecting real-life situations when a model is deployed, and new data is encountered.

3) Rarity Factor Analysis: For each illness, the occurrence rates were analyzed using an empirical method, which is useful in understanding the variations in the illnesses in the dataset. The knowledge of how often each condition occurs helps evaluate how exposed the model is to both low and high-prevalence diseases, thus enhancing balanced performance. As a result of this evaluation of disease prevalence, we can also ensure that the model will not favor categories of diseases that are more common while still being able to detect the scarcer ones, maximizing the real-life uses of the model.

4) Dimensionality Reduction:

- **Correlation Analysis:** In order to detect associations between symptoms, a correlation heatmap was constructed. This helps to remove redundant symptom pairs indicating the presence of features that are highly correlated. In this way, the interplay of these correlations allows one to optimize the model, and in certain cases, reduces dimensionality if such is required.
- **Constant Feature Removal:** When applied to predictive models, features with a variance that is too low do not add any significant value and instead increase the computational resources of the model unnecessarily. With a Variance Threshold set to 0.03, features that have a statistical property of close to zero variance were eliminated from the model to reduce the noise level and improve the efficiency of the model.
- **Mutual Information Analysis:** The contributions of the individual features towards the target variable "Prognosis" were evaluated in terms of Mutual Information (MI) scores. The MI shows the dependence between the feature and the target and assesses how much uncertainty about the target variable is diminished when the value of a certain feature is known, thus capturing the complex nonlinear relations between target and features.

5) Target Analysis:

- **Most Common Diseases:** In order to comprehend the patterns of disease diagnoses, the dataset also contains a visualization of the 10 most common diseases (Fig. 7) in the distribution.

6) Label Encoding: The dependent variable referred to as "Prognosis" is a qualitative variable and thus was label encoded to transform it into a numeric variable. As the dependent variable known as "Prognosis" is a categorical variable it was necessary to encode it using the label encoding technique in order to convert it to a numeric variable. This has been done to allow the machine learning algorithms used to learn the variable and fit the model for effective disease prediction.

C. Disease Classification:

The research will extend to include a set of more experienced machine learning techniques:

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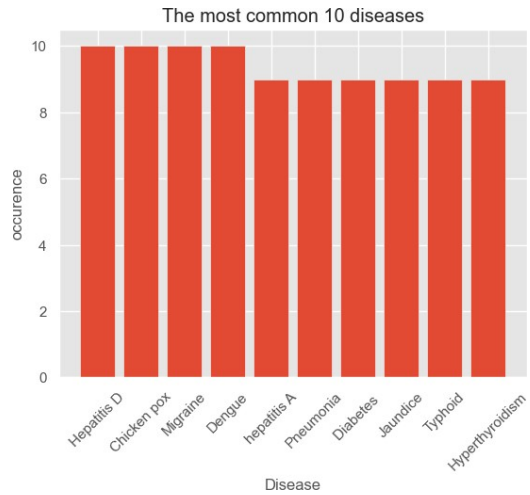

Fig. 7. Top 10 Most Common Diseases

- **Decision Tree Classifier:** An approach that requires hyperparameter tuning and aims for very high accuracy and high interpretability of the model.
- **Bernoulli Naive Bayes:** This approach will be performed with the employment of various feature sets according to their mutual information scores. Such a predisposition to features with higher MI values will improve the predictive capability.
- **Gradient Boosting Classifier:** An effective and robust ensemble approach aimed at achieving a superior performance accuracy through successive learning.
- **Random Forest Classifier:** A tuning process aims at optimizing the parameters of an ensemble technique in order to enhance the reliability of the classification.
- **XGBoost Classifier:** Employing its effectiveness and capability to work with convoluted data structures, with model parameters being optimized for the ultimate end result.
- **Featured Decision Tree:** An model achieved after Feature Engineering using the combination of XGBoost and the Decision Tree.

These advanced algorithms will also be severely tested against a rich set of performance indicators. The latter involves the overall accuracy of the model, the precision of the predictions, its ability to identify relevant cases, and an equally weighted average measure of precision and recall called F1-score.

D. Ayurvedic Medicine Recommendation:

The purpose of this is to suggest or recommend a suitable Ayurvedic medicine after the classification of the disease. The primary aim is to build models such as Decision Tree or Linear Regression for the recommendation. These models will be analyzed to find out which model provides a better and more accurate output.

- **Decision Tree:** Here we will offer understandable suggestions derived from the input features.
- **Linear Regression:** This is used to forecast the effectiveness of therapies by considering continuous characteristics.

Considering the outcomes achieved with these models, the Decision Tree will be singled out as the best because of its ease of understanding and efficient recommendation of Ayurvedic treatments.

V. Results

A. Disease Prediction Accuracy

Different types of algorithms based on machine learning techniques were evaluated for their ability to predict better outcomes. The evaluation involved a decision tree model, a random forest model, a Gradient Boosting Tree, and an implementation of XGBoost.

1) Precision: Precision measures a model's capability of not generating false positive instances. It is determined by the proportion of true positive instances against the total instances defined by the model as positive.

$$\text{Precision} = \frac{\text{True Positive}}{\text{True Positive} + \text{False Positive}} \tag{1}$$

TABLE I
Performance Metrics for Various Models

Model	Accuracy	Precision	Recall	F1 Score
Feature 1 (MI ≥ 0.5)	73.29%	78.57%	74.37%	78.57%
Feature 2 (MI ≥ 0.2)	86.51%	90.48%	87.70%	90.48%
Feature 3 (MI ≥ 0.1)	95.24%	95.24%	94.44%	95.24%
Gradient Boosting	96.72%	94.12%	94.12%	94.12%
Decision Tree	95.78%	94.78%	92.37%	94.62%
Random Forest	97.62%	96.34%	97.56%	96.75%
XGBoost	88.09%	86.18%	90.24%	87.39%
Featured Decision Tree	98.78%	98.78%	98.37%	97.62%

2) Recall: Recall assesses how proficiently a model is able to retrieve every positive instance. It is given by the ratio of the number of positive cases correctly identified to the actual number of positive cases in the data set.

$$\text{Recall} = \frac{\text{True Positive}}{\text{True Positive} + \text{False Negative}} \tag{2}$$

3) F1 Score: The F1 Score is a metric that gauges precision and recall simultaneously. It uses the harmonic mean of the two metrics, making it possible to have a figure measuring the ability of the model in avoiding false positives and false negatives at the same time.

$$F_1 = 2 \times \frac{\text{Precision} \times \text{Recall}}{\text{Precision} + \text{Recall}} \tag{3}$$

Previous studies, like those by Keniya et al. (93.5%) and Reddy et al. (96.1%), focused on high accuracy using algorithms such as KNN and Gradient Boost. The proposed system achieves 98.78% accuracy with a Modified Decision Tree algorithm, prioritizing both performance and interpretability, crucial for healthcare applications.

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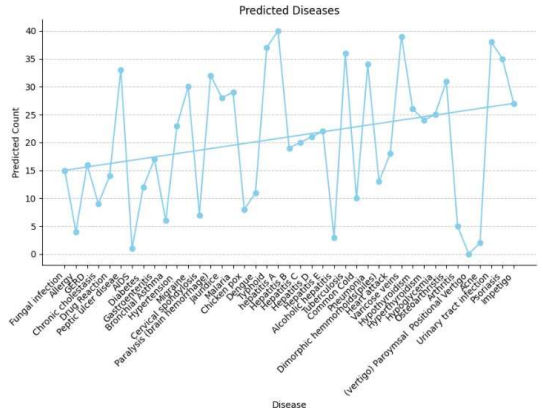

Fig. 8. Illustrating the predicted counts of various diseases, showcasing trends in disease prevalence.

TABLE II
RECOMMENDED AYURVEDIC TREATMENTS BASED ON DISEASE SEVERITY

Disease	Severity	Recommended Ayurvedic Treatment
Migraine	Normal	Jyotishmati
Migraine	High	Vasant Kusumakar Ras
Diarrhea	Low	Pomegranate Drink
Diarrhea	Normal	Lime Juice
Gastritis	Normal	Gulachyadi Kashayam
Gastritis	Low	Laxative
Arthritis	Normal	Nirugandi Oil
Arthritis	Low	Nirugandi Leaves Paste

TABLE III
COMPARATIVE PERFORMANCE METRICS FOR DECISION TREE AND LINEAR REGRESSION

Model	Accuracy	Precision	Recall	F1 Score
Decision Tree (W)	95.35%	95.84%	95.35%	95.28%
Decision Tree (M)	95.35%	69.82%	72.28%	70.19%
Linear Regression (W)	72.48%	68.82%	72.48%	69.86%
Linear Regression (M)	72.48%	45.12%	50.34%	46.71%

B. Ayurvedic Treatment Recommendations

Based on the probable ailments, the model provides individualized treatments grounded in Ayurvedic practices.

These treatments emphasize holistic and natural approaches, tailored to user characteristics such as age, sex, and symptom intensity.

CONCLUSION

Among the models studied in disease classification, the Featured Decision Tree Classifier overshadowed the rest, performing excellently with an accuracy of 98.78%, alongside high precision, recall, and F1 score. This shows that this works well for any classification-related tasks on the given data set. The performance of the Random Forest and Gradient Boosting Classifiers was also commendable achieving accuracies of roughly 97%, but did not go beyond the accuracy of the Decision Tree. The application of the Bernoulli Naive Bayes was encouraging in understanding feature importance but performance-wise lower in comparison.

The Decision Tree model was better than Linear Regression in predicting the Ayurvedic treatment. With an accuracy rate of 95.35% and with better precision, recall, and F1 metrics, the Decision Tree has better trustworthiness and efficacy towards user profile-based recommendations. Therefore, this proves the appropriateness of the Decision Trees in tasks that require classification, such as the recommendations on Ayurvedic medicines, which incorporate the type of constitution of the patient and the severity of the symptoms.

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